Single-Factor Models and Statistical limits of S in Portfolio Optimization

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1 Single factor models

We are tasked to build a Σ with $N = 35$ based on a one-factor model. $\Sigma = \sigma_m^2 \beta\beta^\top + D$ using:

- $B_i \sim U(0.5, 1.5)$
- $\sigma_m = 0.01$ daily
- Idiosyncratic Vol $\sim U(0.01, 0.02)$

1.1 Explanation

A single-factor model mentions that the returns of an asset have two components

1. **A part that comes from the markets:** When the market goes up 1%, almost every asset tends to go up, some more and some less. The magnitude of the asset's reaction is known as beta (β). An asset with a beta of 1.5 amplifies these movements, whereas one with a beta of 0.5 dampens them.
2. **An idiosyncratic component:** It is unrelated to the market. It is idiosyncratic noise and is assumed to be independent across stocks.

1.2 What information do we obtain for Σ?

Recalling that Σ has variances on the diagonal and covariances in the off-diagonal positions:

1. **Covariance:** Between two distinct variables, the only thing they share is the market, specifically, their reaction to it (β). The covariance of these two variables is the product of their two betas and the market variance:

$$\beta_i \beta_j \sigma_m^2$$

2. **Variance:** The variance of a stock consists of the market component plus the variance of its own noise:

$$\beta_i^2 \sigma_m^2 + d_i^2$$

2 Proofs

2.1 S_0 is unbiased, and why $T-1$ is used

We have a set of return vectors $r_1, \ldots, r_T \in \mathbb{R}^N$ that share the same distribution, with:

- Expected value: $E[r_t] = \mu$
- Theoretical covariance matrix: $\Sigma = E\big[(r_t - \mu)(r_t - \mu)^\top\big]$

We define the matrix S_0 as:

$$S_0 = \frac{1}{T} \sum_{t=1}^{T} (r_t - \mu)(r_t - \mu)^\top$$

Proof.

We apply the mathematical expectation to S_0. Due to the linearity of the expectation operator, we can factor out the scalar $\frac{1}{T}$ and move the operator inside the summation:

$$E[S_0] = E\Big[\frac{1}{T} \sum_{t=1}^{T} (r_t - \mu)(r_t - \mu)^\top\Big] = \frac{1}{T} \sum_{t=1}^{T} E\big[(r_t - \mu)(r_t - \mu)^\top\big]$$

Since each return vector follows the same distribution and its expected value is Σ by definition, each term in the summation reduces to Σ:

$$E[S_0] = \frac{1}{T} \sum_{t=1}^{T} \Sigma \qquad \text{(definition of } \Sigma \text{ for each period)} \qquad = \frac{1}{T} \cdot T \cdot \Sigma \; = \Sigma \quad \blacksquare$$

Why $T-1$?

If μ is replaced by $\bar{r}$, calculated using the same data, the deviations $r_t - \bar{r}$ are always smaller. The following identity holds: $\sum_t (r_t - \mu)(r_t - \mu)^\top = \sum_t (r_t - \bar{r})(r_t - \bar{r})^\top + T(\bar{r} - \mu)(\bar{r} - \mu)^\top$. Assuming independent days, the last term equals $T \cdot \Sigma / T = \Sigma$ on average. Consequently, the sum centered around $\bar{r}$ equals $(T-1)\Sigma$ on average: estimating the mean consumes one degree of freedom.

2.2 $\text{rank}(S) \leq T - 1$

Matrix definition of S. Let $X \in \mathbb{R}^{T \times N}$ be the matrix of centered returns, where the rows represent days ($t = 1, \ldots, T$) and the columns represent assets ($i = 1, \ldots, N$), such that $X_{ti} = r_{i,t} - \bar{r}_i$.

Each entry (i, j) of the sample covariance matrix S is calculated as:

$$S_{ij} = \frac{1}{T-1} \sum_{t=1}^{T} X_{ti} X_{tj}$$

This represents the dot product between column i and column j of X, scaled by $T-1$. Therefore, we can express the entire matrix in a compact matrix form as:

$$S = \frac{1}{T-1} X^\top X$$

Given this, we can do the following:

- **1. Invariance under scaling:** Since $T - 1 > 0$ is a constant scalar, multiplying by it does not alter the column space or the rank dimension of the matrix. Therefore:

$$\text{rank}(S) = \text{rank}(X^\top X)$$

- **2. Equality of null spaces and rank:** We analyze the null space of both matrices to see if they share the same dimensionality:
  - If a weight vector w satisfies $Xw = 0$, then it also holds that $X^\top X w = 0$.
  - Conversely, if $X^\top X w = 0$, we can multiply by $w^\top$ to obtain a zero norm:

$$w^\top X^\top X w = \|Xw\|^2 = 0 \implies Xw = 0$$

  This demonstrates that X and $X^\top X$ annihilate exactly the same vectors, meaning they share the same null space. Since both matrices have N columns, by the rank-nullity theorem:

$$\text{rank}(X^\top X) = \text{rank}(X)$$

- **3. Dimensional constraint on centered columns:** By definition, each column of X is centered with respect to its sample mean, which means that the sum of its elements is zero ($\mathbf{1}_T^\top X = 0$). This implies that all columns of X are orthogonal to the vector of ones $\mathbf{1}_T$, forcing them to reside within a vector subspace of maximum dimension $T - 1$. Consequently:

$$\text{rank}(X) \leq T - 1$$

Conclusion regarding the rank. Combining all the preceding steps, we obtain:

$$\text{rank}(S) = \text{rank}(X^\top X) = \text{rank}(X) \leq T - 1 \quad \blacksquare$$

2.3 Show that $E[\lambda_{\max}(S)] \geq \lambda_{\max}(\Sigma)$ and $E[\lambda_{\min}(S)] \leq \lambda_{\min}(\Sigma)$

The result states two things:

- The expected value of the maximum eigenvalue of S overestimates the maximum eigenvalue of Σ.
- The expected value of the minimum eigenvalue of S underestimates the minimum eigenvalue of Σ.

We must define the maximum and minimum eigenvalues of S, for this, we can use the Rayleigh quotient. This quotient describes how much a vector is stretched or shrunk when a matrix is applied to it.

$$R(A, v) = \frac{v^\top A v}{v^\top v}$$

where:

- v is the weights vector
- A is a matrix

We can optimize the quotient to find its maximum and minimum, since we do not want the variance to be artificially inflated, we constrain its norm to 1.

If we perform this optimization, finding the maximum and minimum using the Rayleigh quotient with a norm of 1, we obtain $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$, respectively.

$$\lambda_{\max}(A) = \max_{\|v\|=1} v^\top A v \qquad \lambda_{\min}(A) = \min_{\|v\|=1} v^\top A v$$

The advantage of using the Rayleigh quotient is that it allows us to observe that $\lambda_{\max}$ is convex, since the variance is linear with respect to the matrix, it does not matter how many portfolios are added, if one takes the average across n markets, the portfolio variance will be:

$$v^\top \left(\frac{A_1 + A_2 + \cdots + A_n}{n} \right) v = \frac{v^\top A_1 v + v^\top A_2 v + \cdots + v^\top A_n v}{n}$$

Each term $v^\top A_i v$ cannot be greater than the maximum eigenvalue of the matrix, i.e., $\lambda_{\max}(A_i)$:

$$\frac{v^\top A_1 v + \cdots + v^\top A_n v}{n} \leq \frac{\lambda_{\max}(A_1) + \cdots + \lambda_{\max}(A_n)}{n}$$

Since this inequality holds for any vector v with norm 1, it will also hold for the specific vector that maximizes the left-hand side:

$$\lambda_{\max} \left(\frac{A_1 + A_2 + \cdots + A_n}{n} \right) \leq \frac{\lambda_{\max}(A_1) + \cdots + \lambda_{\max}(A_n)}{n}$$

Since it is convex, we can apply Jensen's inequality, which states that "for a convex function, the average of its values is greater than or equal to the function evaluated at the average of the inputs."

$$E[\lambda_{\max}(S)] \geq \lambda_{\max}(E[S])$$

- $\lambda_{\max}(E[S])$: All covariance matrices are averaged, and the maximum eigenvalue of this average matrix is calculated.
- $E[\lambda_{\max}(S)]$: The maximum eigenvalues of all covariance matrices are averaged.

When all possible covariance matrices are averaged, the extremes of one are smoothed out by the valleys of others, and vice versa, this is why the inequality holds.

Recalling that S is unbiased ($E[S] = \Sigma$), we can rewrite Jensen's inequality as:

$$E[\lambda_{\max}(S)] \geq \lambda_{\max}(\Sigma)$$

In the case of the minimum eigenvalue, we know that the minimum value of a matrix can be viewed as the negative of the maximum value of its opposite matrix:

$$\lambda_{\min}(\Sigma) = -\lambda_{\max}(-\Sigma)$$

Applying this principle to the random matrix $-S$ (whose expectation is $-\Sigma$), and using the previously demonstrated convexity result:

$$E[\lambda_{\max}(-S)] \geq \lambda_{\max}(-\Sigma)$$

If we multiply both sides by -1, which reverses the inequality, we obtain the result for the minimum eigenvalue:

$$E[\lambda_{\min}(S)] \leq \lambda_{\min}(\Sigma) \blacksquare$$

3 D-001 Estimation window and frequency for Σ

3.1 What $q = N/T$ is and why it is the relevant metric.

N is the number of assets, and T is the number of observations. N/T measures the intensity of spectral noise (the extent to which the spectrum of S deviates from that of Σ) within the system being analyzed. This metric is relevant because it allows us to assess how well our sample covariance matrix performs for the purpose of portfolio optimization.

If the dimensionality is low, that is, $N/T \to 0$, we have a sufficient number of observations for each asset, which ensures a robust estimate.

On the other hand, the error grows continuously with q, and if $N/T \geq 1$, we encounter the "curse of dimensionality," where the matrix S is no longer invertible. There are too many variables relative to the small number of observations, since we are attempting to measure

both the behavior of each individual asset and its relationship to the others, meaning there is insufficient information to produce a reliable estimate.

The objective of this part is to compare monthly returns (5 years, $T = 60$) with daily returns (1 year, $T = 252$). For the latter, I will use $T = 250$ as an approximation for the daily returns.

3.2 What does q look like for $T = 60$ and $T = 250$?

1. For $T = 60$, $q = 0.583...$
2. For $T = 250$, $q = 0.14$

3.3 What does each figure show for $T = 60$ and $T = 250$?

Figure 1: It can be observed that for $T = 60$, $\lambda_{\min}$ is underestimated, with $\lambda(S)_{\min}/\lambda_{\min} = 0.1266$. On the other hand, for $T = 250$, $\lambda_{\min}$ is not underestimated as severely, with $\lambda(S)_{\min}/\lambda_{\min} = 0.6591$.

Since the GMV portfolio uses S^{-1}, whose eigenvalues are $1/\lambda$, an underestimated $\lambda_{\min}$ results in a direction that the optimizer perceives as being much less risky than it actually is, consequently, the optimizer assigns a larger share of the weights to that direction.

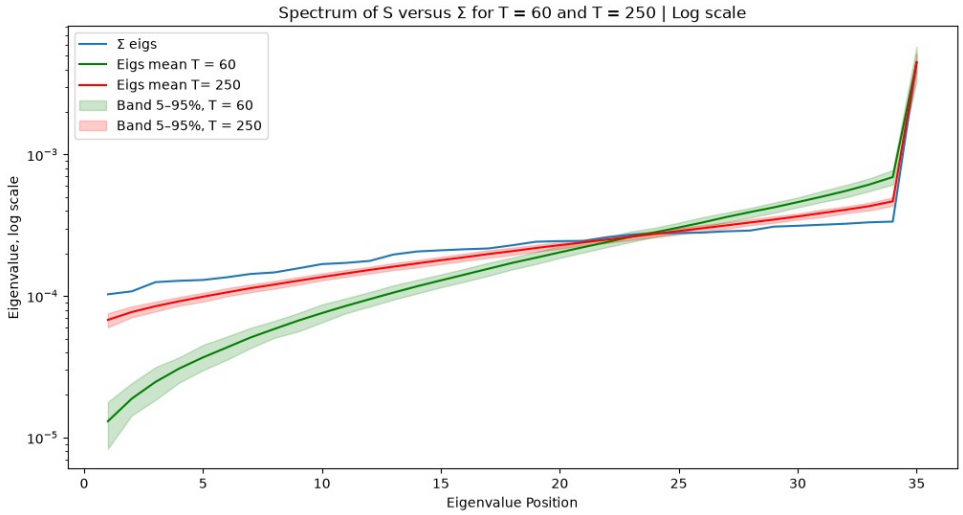
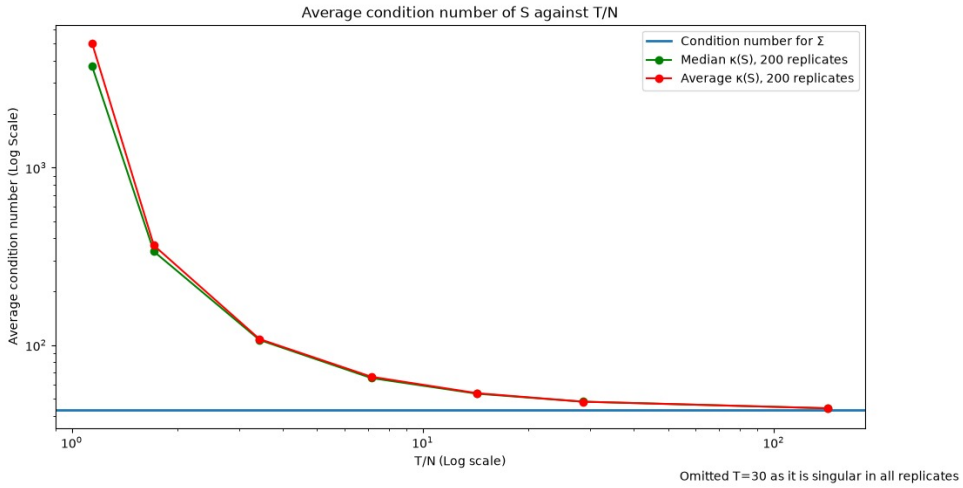


Figure 2: $\kappa(\Sigma) = 43.2417$

1. For $T = 60$, $\kappa(S) \approx 364$. This is 8.4 times larger than $\kappa(\Sigma)$.
2. For $T = 250$, $\kappa(S) \approx 66$. This is 1.5 times larger than $\kappa(\Sigma)$.



The figure shows a downward-sloping curve ranging from $T = 40$ to $T = 5000$. As T increases, the condition number decreases, indicating that the matrix is better conditioned.

3.4 What does the simulation fail to capture?

The simulation only accounts for the noise arising from a limited dataset, it relies on assumptions, such as Σ remaining constant over time and returns being i.i.d. normal, that do not hold in reality. In practice, Σ changes over time, meaning a 5-year window might conflate distinct regimes that are no longer relevant to the present. Furthermore, daily data for illiquid assets tend to result in underestimated correlations and exhibit heavy tails.

3.5 Conclusion

$q = N/T$ measures spectral noise and demonstrates that a daily window ($T = 250$) is superior to a monthly one ($T = 60$) in mitigating the underestimation of $\lambda_{\min}$. This prevents the optimizer from overestimating the reliability of certain directions and keeps the condition number of S close to that of the true matrix Σ. Thus, a larger number of observations significantly reduces the curse of dimensionality and improves the stability of the sample covariance.

However, these simulations assume i.i.d. normal returns and overlook the fact that Σ varies over time, conflating different regimes and underestimating correlations. Therefore, while current tests favor the use of daily data, the choice of window must account for market non-stationarity. Consequently, subsequent analysis will evaluate how shifts in market structure and heavy-tailed distributions affect the performance of the minimum-variance portfolio.